Self Without a Transcript: Semantic Recurrence Across Text and Latent States

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Abstract

What can a frozen language model recover after a description passes through eight hidden-state slots? We compare text relay, latent recurrence and a language roundtrip after the same latent update. Mistral-7B-Instruct-v0.3 and Gemma-2-2B-IT each identified 24/24 literal descriptions across six preference contrasts. Each had 24 distinct updated tensors but one greedy verbalization. In native lexical controls, transplanting downstream middle-layer residuals that had already processed the question reproduced all 12 native preference signs on each checkpoint. We partitioned this span and retested it on six additional topics. On the new topics, a single residual at the period after the second displayed alternative carried 21.8% of the full-span paired margin contrast on Mistral and 27.3% on Gemma. Reversing the donor's option mapping reversed this contrast, while omitting just that position retained 87.9% and 88.8%. These measurements localize an order-sensitive lexical readout at the tested cuts. Question-independent semantic updating and transfer to new tasks remain open.

1 Problem formulation

A description such as "I prioritize keeping decisions revisable" may be reinterpreted as a preference for preserving correction opportunities. The transition may add a useful distinction, remain a paraphrase, or introduce an unsupported claim. Our object is the change in meaning, not the event of uttering the earlier description. We exclude a direct transcript and instructions that infer character from the act of speaking; this does not erase past updates' causal influence on the current state.

We ask whether recurrence yields defensible distinctions, whether a language roundtrip changes their formation or use, and whether changes improve held-out problem solving or appropriate switching under reversed conditions. Convergence is not intrinsically failure, and novelty is not intrinsically success. "Self" denotes an attribution frame rather than a claim about subjective experience.

2 Related work

Coconut trains continuous-state reasoning through hidden-state feedback [5]. Recurrent-depth models study repeated internal computation [4]; STARS motivates distinguishing useful stable dynamics from collapse [13]. These do not establish the validity of arbitrary untrained residual feedback.

Persona vectors and the Assistant Axis connect persona-related behavior with activation structure and interventions [1, 9]. Natural Language Autoencoders provide a learned activation/text

roundtrip [3], while latent chain-of-thought analysis distinguishes output and representational commitment [8]. Our estimand concerns a specific codec rather than "language versus pure thought."

Semantic diversity differs from surface variation [11]. Contrasting self-correction and condition-verification findings motivate operation-specific hypotheses [6, 12]. Dadfar's report is considered with McCardle's open replication critique concerning length, mode and cycle confounds [2, 10]. SlimMoE informs the proposed Phi MoE comparison [7]. We claim neither priority nor a systematic review.

3 Three conditions

Let c_n be a fixed-schedule operation instruction, s_n a current description, and $Z_n \in \mathbb{R}^{K\times d}$ a state. With encoder E, verbalizer D, and state updater G:

$$\text{T:} \quad s_{n+1} = T_\theta(c_n, s_n), \tag{1}$$

$$\text{L:} \quad Z_{n+1} = G_\theta(c_n, Z_n), \tag{2}$$

$$\text{B:} \quad Z_{n+1} = E_\theta(D_\theta(G_\theta(c_n, Z_n))). \tag{3}$$

L and B begin from copies of $E(s_0)$ and share weights, boundaries, transport, generation policy and schedule. Their contrast is primary; T/L is a system-level comparison.

Each outer step uses a fresh context without persistent KV cache. Only current text or slots carry forward. Observation uses clones and restores RNG state; its text is not fed back except as prescribed by B. The initial description is not repeatedly reintroduced. Slots occupy explicit decoder boundaries; post-writer to pre-cut transport is explicit. An identity map is an unqualified engineering diagnostic.

The two schedules are an ordinary meaning update and a fixed sequence of decomposition, boundary examination and reintegration. Unchanged content is allowed when no justified update exists. Adaptive cycle-triggered feedback would be a separate condition.

4 Qualification and measurement

Native parity, content discrimination, contrastive-state swap effects, one-step validity, readout fidelity and positional controls precede any latent pilot. Transport or codec training may be needed; training material must remain disjoint from confirmation. One-step success does not establish long-trajectory validity.

Meaning. Blind paired judgments assess support, update type, new distinctions and event-recovery violations. Lexical and tensor distances are not semantic scores.

Activations. Selected residual, attention, MLP and supported MoE gate outputs are retained. Gate logits are not actual dispatch frequencies. The probes below test source discrimination and intervene on residual positions.

Effects. Problems arrive only after recurrence and operate on isolated copies. Near, structural and less-related transfer are separate tiers. Initial-state, one-shot instruction and non-self controls are planned. Reversals distinguish commitment from rigidity. Frozen-weight state-conditioned gains are not permanent learning.

At a shared state Z, one branch continues with $G^r(Z)$ and another with $G^r(E(D(Z)))$. The development probes below add slot order and magnitude controls. Compression matched for distance, no-update roundtrips and independent semantic review remain open.

5 Development design

The first experiment compared native EOS with an eight-token ceiling on two Mistral descriptions about revisable or prompt decisions. We subsequently restored the archived literal, encoded, latent and roundtripped states and asked which priority each donor favored. Before collecting new scores, we fixed six additional topics with two opposing preferences and two phrasings per preference. They covered filing order, walking routes, meeting notes, temperature units, shelving and directions. For each topic all donors received the same two alternatives. The question and alternatives were absent during encoding, the single plain update and greedy verbalization.

For each donor we scored both answer labels and reversed their option assignments. The plus margin is the mean, over the two orders, of the log probability of the label naming the plus option minus that of the label naming the minus option. A positive margin favors the plus option. A pair is ordered if the plus donor has a larger margin than the minus donor. Correct identification of both donors additionally requires the two margins to straddle zero in the expected direction. Ties use the stored scores at their recorded precision. Each checkpoint contributed 12 matched pairs in six topic clusters. All results are descriptive. Alternative orders and phrasings are repeated measures.

We scored literal text, E, G after one update, B after re-encoding, reversed G rows, and G vectors with the donor's per-slot magnitudes on fixed directions. Zero vectors and uninjected native placeholders served as null inputs. For each of 24 descriptions we collected D(E) and D(G), and we generated from both nulls once. Both checkpoints used one RTX 5090, eager BF16, eight slots, uncalibrated identity transport, a common 1024-token context and their native EOS ids. The cut and writer were layers 16 and 31 for Mistral and layers 13 and 25 for Gemma (zero-indexed). There was no training, prompt selection against these scores or human semantic rating.

6 Results

6.1 Native EOS and the language roundtrip

Two opposed descriptions emphasized revisable or prompt decisions. Each underwent two plain updates in T, L and B, once with native EOS and once with an eight-token ceiling. The four trials used greedy BF16 Mistral-7B-Instruct-v0.3, eight slots, a 1024-token context, a pre-layer-16 cut and a post-layer-31 writer (zero-indexed). All 18 generations per policy include main-path, observer and branch readouts. All 18 reached EOS with the full context available, and all 18 reached the explicit ceiling under the cap. With EOS, each of the four B updates verbalized the same 13-token sentence about workspace contents. The four B states matched within each policy, although the two policies gave different states. The four L states differed within each policy and matched step for step across policies.

6.2 Reading the archived state donors

For the previously studied Mistral pair, the literal revisability and speed descriptions yielded plus margins of +16.766 and -16.344 nats. The two initial slot states both gave +2.844. The first and second L updates gave distinct fingerprints but positive margins for both donors. B gave +2.813

for both donors at both steps. This particular question separated the two literal descriptions and favored revisability for every archived latent donor.

6.3 Six contrasts and matched controls

On both checkpoints all 24 literal descriptions had the correct sign. At E and G, none of the 12 matched pairs had both signs correct on either model. The B margins tied in all 12 pairs on each model. Each checkpoint had 24 distinct E tensors and 24 distinct G tensors, but just one E readout string and one G readout string per checkpoint and stage. Re-encoding D(G) produced one B tensor on each checkpoint. All 50 free generations per checkpoint ended at native EOS. Gemma's G readout requested a description to summarize, as did its zero-state readout. Mistral's G readout referred to the contents of the work areas.

Table 1: Ordered/tied pole pairs (of 12) and correct donor signs (of 24). Each pair shares its question and counterbalanced labels.

Route	Mistral pairs	Gemma pairs	Mistral signs	Gemma signs
Literal	12/0	12/0	24	24
E	6/2	5/1	12	12
G	3/0	7/1	12	11
B	0/12	0/12	12	12
G, reversed	5/2	5/0	12	13
G, magnitude only	9/0	3/2	12	12

The row-reversal and fixed-direction magnitude shams produced some positive rankings even when most donor signs were unchanged. The absolute magnitude of a state's response matters for interpreting these rankings. For ordinary G the median signed pole gap was -0.016 nats on Mistral and approximately zero on Gemma, compared with +29.680 and +20.766 for literal text. Per-topic opposite-pole and same-pole paraphrase differences, exact ties and both raw label orders are included in the supplement.

6.4 Matched lexical donors and replay geometry

A lexical positive control placed the words of one answer option at the eight designated positions, padded on the right with the neutral period token. No answer letter was included. We compared native scoring, ordinary E on the same words, and injection of residuals captured at the same positions in a prefix ending at the last slot. Native words yielded 11/12 correct donor signs on Mistral and 12/12 on Gemma. Ordinary E gave 6/12 on each. The short-prefix aligned donor gave 6/12 and 7/12. Replaying neutral slots captured from a shorter forward changed individual label log probabilities by up to 0.1903 and 0.1613 nats. Those discrepancies motivated a separate length-matched experiment.

We appended one neutral dummy token to every scoring prompt and read the logit vector immediately before it. This reproduced all earlier native label scores exactly. The eight-slot donor now used a full-length forward, with neutral tokens replacing the question and alternatives after the slots. Neutral replay then matched every native vocabulary logit in all 12 question/order combinations on both checkpoints. The opaque-tail lexical donor's slots also matched its natural full-length counterpart exactly in all 24 donor/orders per checkpoint.

Table 2: Correct donor signs and median signed pole gap in nats. The same six paired option phrases are used for each intervention.

Intervention	Mistral correct	Gemma correct	Mistral gap	Gemma gap
Native words	11/12	12/12	28.219	19.672
Eight slots	6/12	7/12	0.008	0.219
Downstream span	11/12	12/12	27.594	19.734
Both regions	11/12	12/12	28.219	19.672

We next transplanted the natural lexical donor's middle-layer residuals from the positions after the slots through the scoring position. These downstream residuals had already processed the question and alternatives. Transplanting only this region into a neutral recipient reproduced the native preference sign in all 12 cases per checkpoint, including Mistral's error on the author-shelving donor. Its full logit vectors could still differ. Transplanting the slots and downstream region together recovered the entire native scoring-position logit vector exactly in 24/24 donor/orders per checkpoint.

Supplementary plans, data and analyses.

6.5 Localizing the downstream span

We split the interval from the end of the slots through the scoring position into the generic task instruction, topic question, labeled alternatives, closing text before scoring, and the scoring position. Native tokenizer offsets were checked against the exact recipient IDs. We transplanted each block alone into a neutral recipient, left it out of an otherwise complete downstream transplant, and replaced it with neutral residuals in a native donor. We also tested windows of equal width. The six original topics and six prospectively fixed new topics used the same lexical task, both label orders and both donor poles. Each checkpoint and panel provided 24 donor/order measurements per study. The two panels are reported separately.

For each topic, let the pole gap be the plus donor's order-averaged margin minus the minus donor's margin. The reported percentage is 100 times the sum of those six gaps under a given intervention divided by their sum under whole-downstream transfer. A negative value indicates a reversed aggregate direction. Only-window and leave-window-out values need not add to 100%.

Transplanting the labeled alternatives alone yielded 49.5% to 84.8% of the full-span contrast across the four checkpoint/panel combinations. Omitting them left 4.8% to 22.9%. On the new panel the last eight positions retained 90.8% on Mistral and 8.7% on Gemma. In the Mistral template the closing span is six tokens long, so the last eight include two option positions. In Gemma the closing span is ten tokens long, so the last eight contain no option token.

Table 3: Percentage of the full downstream pole gap. Each window is transplanted alone except in the omission column.

Panel	Model	Options	Omit options	Score	First 8	Last 8
Original	Mistral	70.4	20.4	20.7	0.0	90.6
Original	Gemma	84.8	4.8	1.6	-0.1	5.7
New	Mistral	49.5	22.9	26.6	0.0	90.8
New	Gemma	79.3	8.4	0.6	0.2	8.7

6.6 What reaches the period after the second alternative?

The original Mistral last-eight result motivated the option-clause follow-up, fixed before inspecting prospective-panel outcomes. Both clause-ending periods were separate from the phrases and had the same token ID within each model. On new topics the first period alone carried 9.2% and 11.9% of the full contrast on Mistral and Gemma, respectively. The second period carried 21.8% and 27.3%. Omitting this period retained 87.9% and 88.8%. Leaving one topic out gave ranges of 18.5% to 23.9% and 25.6% to 28.4% for the second-period-only measurement.

Table 4: Percentage of the whole downstream pole gap at the second displayed option's period. Context columns transplant the single period into the real recipient. The omission column keeps all other downstream positions from the matched donor.

Panel	Model	Matched	Hide stem	Hide phrases	Reverse	Omit period
Original	Mistral	37.8	44.5	0.7	-32.9	76.6
Original	Gemma	38.8	31.5	0.2	-33.3	82.2
New	Mistral	21.8	16.4	0.0	-22.3	87.9
New	Gemma	27.3	20.7	0.0	-26.4	88.8

We held the target's true wording and option order fixed. Masking only the donor's question stem with neutral tokens of the same length preserved a positive contrast at the second period. Masking the two option phrases while retaining their printed labels and separators reduced the absolute raw contrast below 0.7% of the full reference in all four cells. Reversing the donor's natural option assignment changed the aggregate sign in every cell, including after subtracting the source neutral background and adding the lexical difference to the target background. On the new panel, among individual orders where the donor-native and target-native preferred letters differed, the raw second-period transplant followed the donor letter in 19/22 Mistral and 19/24 Gemma cases. The first period sometimes shifted under order reversal and was excluded whenever its absolute position or token ID differed. The second period was compatible in all 24 donor/orders per cell.

All 192 donor/order records across the eight jobs passed exact neutral and combined slot-plus-suffix vocabulary-logit replay. Original-panel native and whole-suffix label scores reproduced the earlier experiment exactly. The computations used no free generation.

Supplementary localization plans, likelihoods and per-topic results.

7 Interpretation and limitations

In these fixtures the complete downstream region reproduced the native lexical choices. The partition and masking experiments narrow this observation to an option-order-sensitive contribution at the second displayed alternative's terminal period. Its counterbalanced contrast persisted with the question stem hidden, weakened when the option words were hidden, and reversed under a natural option-order swap. The large contrast retained on omission leaves room for other positions and their interactions.

The second period is later in the causal sequence and has access to both alternatives. These interventions have not separated separator identity from order, position or preceding lexical context. Neutral-token masks alter the donor distribution. The two panels share the same explicit lexical schema, and each checkpoint was tested at one middle-layer cut in BF16. The short-prefix replay discrepancy was removed by matching forward length, but its numerical kernel has not been identi-

fied. Question-independent state portability, independent evaluation of one-step meaning updates, blind judgments of new distinctions and held-out transfer remain open. No semantic gate has been approved.

8 Proposed pilot and analysis

Six content groups, two poles, two paraphrases and self/other frames yield 48 development seeds. Three arms and two schedules yield 288 proposed trajectories per model. Candidate families are Mistral-7B-Instruct-v0.3, Llama-3.2-3B-Instruct, Gemma-2-2B-IT and Phi-mini-MoE, with Phi-tiny-MoE a separately reported fallback. Family differences are not controlled architectural interventions.

Twelve steps with fixed assessment checkpoints are proposed. The present CLI scores tasks at baseline/final only. Its 20 synthetic tasks are exposed development fixtures, not a broad benchmark. Confirmation data, thresholds, sample size and preregistration are pending.

Paired L/B utility differences at a declared depth will be reported alongside support and switching. Inference must account for dependence among meaning groups and task families rather than treating each step as independent. Same-step designs report actual compute; matched-compute analyses are secondary. Failures, censored generations and ties retain denominators. Best-depth selection on confirmation is prohibited.

References

[1] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL `https://arxiv.org/abs/2507.21509`.

[2] Zachary Pedram Dadfar. When models examine themselves: Vocabulary-activation correspondence in self-referential processing, 2026. URL `https://arxiv.org/abs/2602.11358`.

[3] Kit Fraser-Taliente, Subhash Kantamneni, Euan Ong, Dan Mossing, Christina Lu, Paul C. Bogdan, Emmanuel Ameisen, James Chen, Dzmitry Kishylau, Adam Pearce, Julius Tarng, Alex Wu, Jeff Wu, Yang Zhang, Daniel M. Ziegler, Evan Hubinger, Joshua Batson, Jack Lindsey, Samuel Zimmerman, and Samuel Marks. Natural language autoencoders produce unsupervised explanations of llm activations. *Transformer Circuits Thread*, 2026. URL `https://transformer-circuits.pub/2026/nla/index.html`.

[4] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach, 2025. URL `https://arxiv.org/abs/2502.05171`.

[5] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space, 2024. URL `https://arxiv.org/abs/2412.06769`.

[6] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *The Twelfth International Conference on Learning Representa-*

tions, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/8b4add8b0aa8749d80a34ca5d941c355-Abstract-Conference.html.

[7] Zichong Li, Chen Liang, Zixuan Zhang, Ilgee Hong, Young Jin Kim, Weizhu Chen, and Tuo Zhao. SlimMoE: Structured compression of large MoE models via expert slimming and distillation, 2025. URL https://arxiv.org/abs/2506.18349.

[8] Zirui Li, Xuefeng Bai, Kehai Chen, Yizhi Li, Jian Yang, Chenghua Lin, and Min Zhang. Dynamics within latent chain-of-thought: An empirical study of causal structure. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 65859–65883. PMLR, 2026. URL https://proceedings.mlr.press/v306/li26c.html.

[9] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 80955–81003. PMLR, 2026. URL https://proceedings.mlr.press/v306/lu26g.html.

[10] John P. McCardle. [¬Re] vocabulary-activation correspondence in self-referential llm processing, 2026. URL https://github.com/ReScience/submissions/issues/105. Replication report submitted to ReScience; submission open and under review at the 2026-11-01 audit. The title retains the author's negated-Re marker.

[11] Yiheng Tao, Kaiwen Cheng, Zhiwei Nie, Chang Liu, and Jie Chen. Divergent thinking: Escape the homogeneity trap in generative commonsense reasoning. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 18395–18410. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.915. URL https://aclanthology.org/2026.findings-acl.915/.

[12] Zhenyu Wu, Qingkai Zeng, Zhihan Zhang, Zhaoxuan Tan, Chao Shen, and Meng Jiang. Large language models can self-correct with key condition verification. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 12846–12867. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.714. URL https://aclanthology.org/2024.emnlp-main.714/.

[13] Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, Wen-Da Wei, Jie-Jing Shao, Lan-Zhe Guo, and Yu-Feng Li. Stabilizing recurrent dynamics for test-time scalable latent reasoning in looped language models, 2026. URL https://arxiv.org/abs/2605.26733. arXiv metadata lists ICML 2026; arXiv version cited here.